INCH-POUND

MIL-I-27677E(USAF)
18 May 1990
SUPERSEDING
MIL-I-27677D(USAF)
7 March 1966

MILITARY SPECIFICATION

INDICATOR, LIQUID QUANTITY OXYGEN

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for liquid oxygen quantity indicators. The indicators are of the capacitance type and indicate from 0 to 75 liters of liquid oxygen.

1.2 Classification. The indicators shall be of the following types, as specified (see 6.2):

GMU-34/A	Cover glass coated with anti-reflection coating in accordance with MIL-C-675.
GMU-34A/A	Cover glass coated with high efficiency anti-reflection coating as defined herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)."

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated - Plywood.
PPP-B-636	Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Support Division, San Antonio ALC/MMDEM, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6680

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MILITARY

MIL-P-116	Preservation, Method of.
MIL-C-675	Coating of Glass Optical Elements (Anti-reflection).
MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification for.
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for.

STANDARDS

FEDERAL

FED-STD-595	Colors.

(Requirements for individual color chips, 3 X 5 supplements should be submitted to the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods.
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D-3951	Packaging, Commercial.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25645 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Configuration. The indicator shall consist of a single unit containing all of the equipment necessary to meet the performance requirements specified herein and in MIL-I-25645.

3.4 Case. The length of the case shall not exceed 5.0 inches.

3.5 Dial. The dial for the indicator shall conform to figure 1.

3.6 Cover glass coating (GMU-34A/A). The cover glass for the GMU-34A/A indicator shall be provided with a reflection-reducing coating which meets the requirements of MIL-C-675 except that the following reflectance tolerances shall apply:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0°	450 - 675	0.6 maximum
0°	425 - 700	0.5 average
30°	450 - 625	1.0 maximum
30°	425 - 700	0.5 average

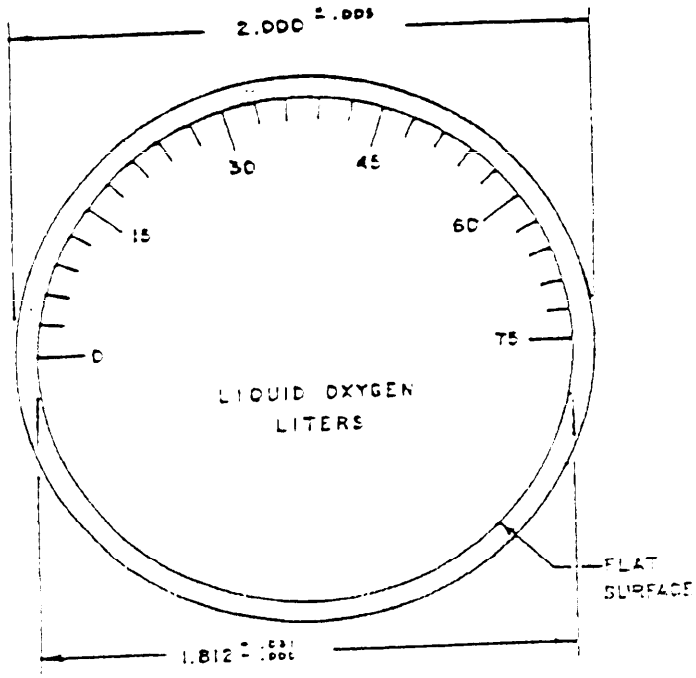
3.7 Electrical connector. The electrical connector shall be in accordance with figure 2. The indicator shall be designed internally so that it will function properly when the mating electrical connector is wired as shown on figure 2. The coaxial shield shall be grounded within the indicator.

3.8 Capacitance. For design purposes, the capacitance for zero reading shall be 910.0 for micromicrofarad (uuf).

3.9 Adjustments. The zero calibration adjustment shall be capable of adjusting the pointer to zero indication over an input capacitance range of at least 900.0 to 950.0 uuf. With the indicator adjusted to zero indication at 900.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 1335.0 uuf input capacitance by means of the full calibration adjustment. With the indicator adjusted to zero indication at 950.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 1385.0 uuf capacitance by means of the full calibration adjustment.

3.10 Weight. The indicator shall weigh not more than 1.5 pounds.

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NUMERALS AND LETTERS SHALL CONFORM TO MS33558

MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	FINISH COLOR PER FED - STD-595
Numerals	0.140	0.025	
Liquid Oxygen	0.140	0.020	
Major Grad.	0.188	0.031	37875 (Lusterless White)
Minor Grad.	0.125	0.015	
Liters	0.100	0.015	
Background of Dial	—	—	37038 (Lusterless Black)

DIMENSIONS IN INCHES

FIGURE 1. Dial.

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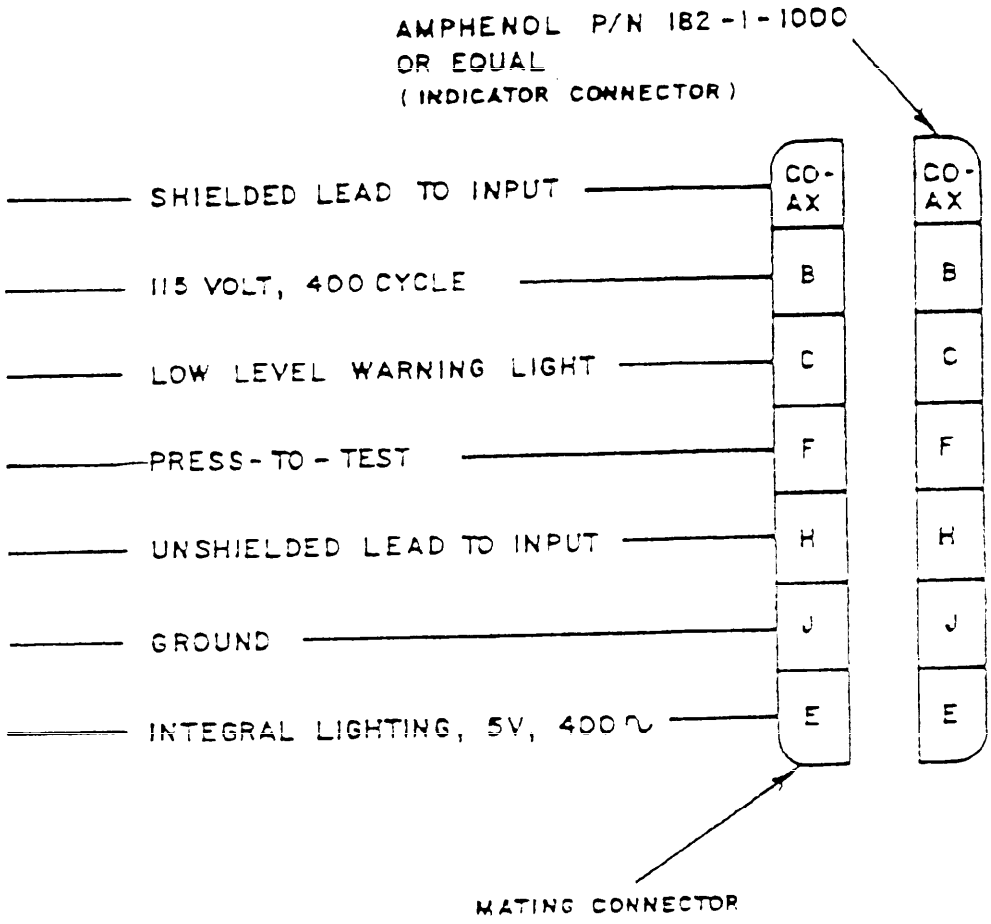


FIGURE 2. Connector wiring.

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3.11 Accuracy. When the indicator is tested as specified in MIL-I-25645, the accuracy of the indicator shall be in accordance with table I. The tolerance for room-temperature readings is the allowable deviation from the nominal calculated values. All other tolerances are the allowable deviations from the actual room-temperature readings.

TABLE I. Tolerances for Capacitance Readings.

Test	Tolerances plus or minus
Accuracy - room temperature	5.4 uuf
Low temperature operation	7.6 uuf
High temperature operation	7.6 uuf
Vibration	4.3 uuf
Power supply variation	4.3 uuf
Low temperature storage and operation	7.6 uuf
Salt spray	4.3 uuf
Cycling	4.3 uuf

3.12 Integral lighting. The indicator shall have integral lighting conforming to MIL-L-27160.

3.13 Recycled, reclaimed, and virgin materials. Except when intended use of the item will be jeopardized by the use of reclaimed or recycled materials, in preparing new and revising or amending specifications, preparing activities will ensure that:

a. There is no exclusion to the use of recovered materials.

b. There is no requirement that an item be manufactured from virgin materials.

c. Within 1 year from the date of issue by the Environmental Protection Agency of guidelines designating items which are or can be produced with recovered materials, specifications for such products require the use of recovered materials to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, inspection, and testing procedures. Sampling, inspecting, and testing procedures shall be in accordance with MIL-I-25645.

4.3 First article inspection. The first article test shall consist of all tests described under test methods of MIL-I-25645, including reliability and longevity tests which shall be conducted in accordance with MIL-STD-781. Two GMU-34A/A indicators shall be subjected to the additional test described in 4.4.1.

4.4 Test methods.

4.4.1 Cover glass reflectance (GMU-34A/A). The reflectance of the GMU-34A/A indicator cover glass shall be measured and shall be within the tolerances specified in 3.6.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A, C or commercial/industrial as specified (see 6.2).

5.1.1 Level A. The indicator shall be preserved and packaged in accordance with method 1C of MIL-P-116.

5.1.2 Level C. The level C preservation shall conform to the requirements of this level.

OR

Commercial/industrial preservation of indicator shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing.

5.2.1 Level A. The indicator preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-601, overseas type. The closure shall be in accordance with the appendix of the container specification.

5.2.2 Level B. The indicator preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-636, weather resistant. The closure shall be in accordance with the appendix of the container specification.

5.2.3 Level C. The indicator preserved and packaged as specified in 5.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Container shall be in accordance with either the uniform freight classification rules or regulation of other carriers applicable to the mode of transportation.

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or

mercial/industrial packing of indicator shall be in accordance with the uirements of ASTM D 3951.

5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing ll be in accordance with MIL-STD-1186. Waterproofing requirements cushion- material and container shall be waived for domestic shipments. The drop ts of MIL-STD-1186 shall be waived when the items are preserved and packaged immediate use or when the drop tests MIL-P-116 are applicable.

5.4 Marking.

5.4.1 Level A, B, C. In addition to any special or other identification king required by the contract (see 6.2) each unit pack intermediate and erior container and utilized lead shall be marked in accordance with -STD-129.

5.4.2 Commercial/industrial marking shall be in accordance with the uirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The indicators covered by this specification are ended for use in indicating the quantity of liquid oxygen remaining in con- ters in C-5A and C-141 airplanes.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of indicator required (see 1.2).

c. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial.

d. First article inspection and testing, if required (see 3.1 and 4.3).

6.3 First Article. When a first article inspection is required, the item be tested shall be a first article sample or it may be a standard production m from the contractor's current inventory. The first article shall consist 2 units. The contracting officer shall include specific instructions in uisiton documents regarding arrangements for examinations, tests and approval the first article.

6.4 Subject term (key word) listing.

Indicator
Liquid Oxygen
Cover Glass
Dial

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 6680-F207)

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